Click Prompt Learning With Feature Encoding for Segmentation of Remote Sensing Images

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***Abstract*— Pixel-level annotation tasks are important in the intelligent processing of remote sensing images. For these tasks, interactive image segmentation (IIS) models using click prompts are developing fast in the field of natural images. However, most interactive segmentation models using click prompts are unsuitable for remote sensing images with their current design of click prompts and their interaction schemes with image information. Based on the situation, we used a DETR-like model as the basic framework and redesigned the pixel decoder and the transformer decoder to better suit the task of IIS for remote sensing images. In the pixel decoder, we designed a click prompt with feature encoding to learn click information and a composite attention structure to facilitate interaction between click and image information, allowing the image feature at the click locations to more easily dominate annotation masks. In the transformer decoder, we utilized deformable attention, using only a single initialized query to obtain annotation masks and IoU prediction. In this article, we trained our model on a composite remote sensing dataset and evaluated its performance on external datasets. The results showcased the model's adaptability, achieving superior performance compared with existing methods. The code will be available at: https://github.com/songbingze/ClickPromptRSIIS**

***Index Terms*— Click prompt learning, interactive image segmentation (IIS), remote sensing image, transformer.**

I. Introduction

PIXEL-LEVEL annotation tasks are important in the intelligent processing of remote sensing images, forming the cornerstone for various deep learning algorithms, such as semantic segmentation [1], [2], [3], instance segmentation [4], [5], and change detection [6], [7]. Although extensive research has been conducted on classification and segmentation methods in the field of automatic annotation [8], [9], [10], many practical applications continue to demand exceptionally high precision. Consequently, these applications still heavily depend on manual annotation or manual corrections supplemented by basic image processing algorithms. If models trained on existing datasets perform inadequately, we often need to reannotate sample data to improve their performances [11], [12], [13]. One of the key bottlenecks in achieving intelligent remote sensing information processing lies in obtaining annotated data.

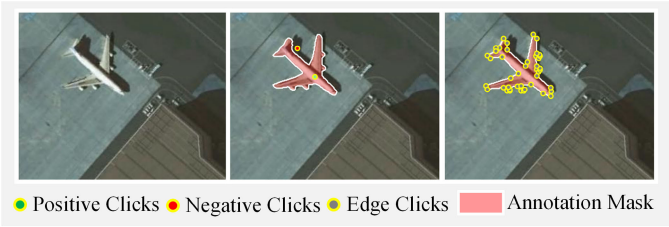


Fig. 1. (Left) Original image. (Middle) Annotated result of IIS with click prompts.

However, the diversity in imaging scales, modalities, environments, and application-specific requirements of remote sensing satellites presents substantial challenges for pixel-level annotation tasks [14]. Traditional image annotation methods often struggle to maintain consistency and quality across datasets, while being time-consuming and labor-intensive. In recent years, interactive image segmentation (IIS) enhanced by deep learning aims to rapidly generate accurate binary annotations through user interaction, thereby obtaining high-quality pixel-level annotations [15]. Users can provide interaction in various forms, including clicks [16], [17], [18], bounding boxes [19], [20], polygons [21], [22], scribbles [23], [24], [25], text [26], or combinations thereof [27]. Among these, click-based annotation algorithms are notable for their simplicity and ease of use, allowing for convenient object annotation, as shown in Fig. 1.

The advancement of IIS with click prompts in remote sensing has been slow, mainly due to the lack of large, multicategory datasets needed to train models with strong generalization. In addition, the scarcity of high-quality annotated instance segmentation datasets further compounds this challenge. Some studies train interactive segmentation models on semantic segmentation datasets, enabling click-based annotation of all objects in the same category [28], [29],

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[30]. However, this approach typically involves training on only a single-scene remote sensing segmentation dataset, further decreasing the generalization ability of the obtained model. Meanwhile, some studies have explored training an IIS model on natural image datasets to annotate remote sensing images [26], [31]. These models exhibit relatively good annotation performance on high-definition and texture-uniform remote sensing objects. In practical remote sensing annotation, factors like image quality and resolution reduce the efficiency of these models compared with those specifically trained on remote sensing data. As a result, IIS with click prompts in remote sensing remains limited by these constraints, underscoring the need for comprehensive datasets to train models with robust generalization to overcome these challenges.

In addition, in natural image processing, IIS models using click prompts have achieved significant progress. Researchers have explored various aspects of IIS using click prompt tasks, including the design of click prompt [17], [26], [32], interactive segmentation networks [18], [33], [34], the generation of click during training [17], [35], postprocessing techniques [15], [36], loss functions [17], [37], and dynamic backpropagation [15], [37]. These studies have demonstrated the potential of IIS with click prompts for rapidly annotating regions of interest in natural images.

Research on IIS with click prompts has focused on encoding click prompts, their interaction with images, and designing IIS networks within Detection Transformers (DETRs)-like [38] model frameworks.

Regarding the encoding click prompts, current IIS models represent click prompts using learnable click label and position encodings [26], [32]. However, this method only captures label and position information and lacks the ability to convey semantic information, limiting its effectiveness in processing image features. Learnable encodings also pose training challenges. Therefore, we have designed a click prompt that directly extracts features from the image, incorporating semantic, positional, and label information at the clicked location. This comprehensive approach enhances annotation accuracy.

Regarding the interaction between click prompts and images, there are currently three methods for interacting. One is to directly sum the features of the click map features and the image features [11], [34], as shown in Fig. 2(a). Although this method is simple, it has low interaction efficiency, requires deep network processing, and usually needs to be input at the head. The second method is to use attention for both the click map and the image [33], [37], as shown in Fig. 2(b), which requires a large amount of computation. The third form is to directly apply attention to the click prompt and the image features [26], [32], as shown in Fig. 2(c), which requires less computation and has high interaction efficiency. This approach is computationally efficient but tends to only consider features at the click location, leading to potential errors at object edges. Therefore, we have designed a composite structure, as shown in Fig. 2(d), that first focuses attention on the pixels near the click to obtain information from the image, and updates the click prompt. Then, we allow this click prompt to correct the features of the entire image, enhancing the accuracy of the annotation mask.

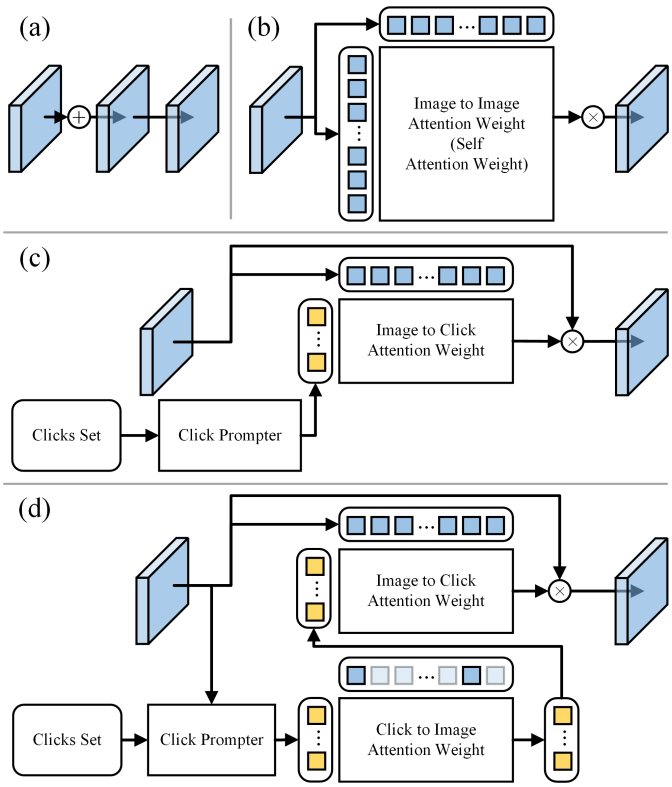


Fig. 2. Click-image information feature fusion structure. (a) Sum of the click features and image features. (b) Image self-attention. (c) Cross attention of clicks prompts to the image feature. (d) Compound attention we propose.

Regarding designing the IIS task within DETR-like model frameworks, image self-attention is primarily used in the pixel decoder to obtain image features, while click prompts are input into the transformer decoder [32], [37]. This setup only affects image features globally, making it difficult to influence features locally with click information. We propose inputting click prompts directly into the pixel decoder, allowing each pixel to focus on the prompts, which we believe is more effective for IIS tasks. In addition, since IIS produces binary segmentation results, current methods often design the transformer decoder similar to multiclass semantic or instance segmentation tasks, leading to cumbersome training and homogenized queries. We address this by simplifying the transformer decoder to use a single initialized mask query for annotation and IoU predictions.

To summarize, our contributions are as follows.

1) Utilized a combined dataset of common instance and noninstance categories in high-resolution remote sensing imagery to train a highly adaptable IIS model.
2) Designed an encoding method for click prompt learning based on image features, allowing the click prompt to convey more comprehensive information and improving annotation accuracy.
3) Designed a composite attention structure that enables the click prompt to focus on surrounding information at each scale, enhancing representation accuracy and facilitating more precise mask feature learning.

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4) Simplified the transformer decoder by using a single initial query for mask feature attention and IoU prediction, resulting in a simpler structure with reduced computational complexity.

II. Related Works

A. Interactive Segmentation Methods

1) Nondeep Learning Methods: Early nondeep learning methods typically segmented objects based on boundary and region properties. Live Wire [39] and Intelligent Scissors [40] require placing clicks on the boundary to annotate the object region. Random Walk [24], Graph Cuts [23], and Region Merging [41] require the user to annotate clicks or scribbles in the foreground and background regions. GrabCut [42] should be initialized with a bounding box or scribble to segment the object. Such methods can only capture shallow features of the image, thus relying on statistical learning methods to calculate boundary and region features to obtain segmentation results. When dealing with complex objects, they often struggle to achieve satisfactory segmentation results.

2) Deep Learning Methods: Deep learning algorithms for interactive content can be divided into two categories based on the type of interaction: only clicks and mixed interaction. Click-only method typically refers to the annotator inputting positive and negative clicks to the model. Mixed interaction typically includes one or several of the following annotation methods: points, scribbles, bounding boxes, and text.

Among them, click-only models are the most well-developed due to their simplicity and serving for automated annotation. Deep interactive object selection (DOS) [35] discusses how to automatically generate clicks during the training process. It has also designed the basic framework for an IIS model. The BRS [16] and f-BRS [15] enable the model to update parameters during the interactive process. In BRS, all parameters are updated during the annotation process, while f-BRS only updates a subset of these parameters. RITM [17] trains the model by using larger datasets with finer annotations. Besides, during training, RITM dynamically generates clicks in mislabeled areas to simulate the human annotation process, resulting in more accurate annotations. FocalClick [36] accelerated the inference speed by predicting and updating segmentation results in local regions, allowing the model to work on smaller input image sizes. Pseudoclick [43] predicts the possible location of the next click during the prediction process and uses it as input. Subsequently, SimpleClick [34] became the first to adopt vision transformer (ViT) [44] in the backbone, marking the transition of IIS models into the Transformer era. Later, IIS models also see improvements using DETR-like model (a Transformer-based framework, which will be detailed in Section II-B). PiClick [32] improves the transformer decoder from the DETR [38] to achieve more accurate multioutput results. Also, it adopts a learnable click embedding design. AdaptiveClick [37] integrates the design of click maps into a DETR-like model and makes improvements to the loss function.

Regarding remote sensing images, DRE-Net [30] trains on urban segmentation datasets and designs a flexible click map generation method. ARE-Net [45] has designed a specialized annotation model specifically for buildings and has devised a two-stage training approach.

Mixed interaction methods aim to obtain more accurate human priors through diverse interaction modes. DEXTR [46] requires placing annotation clicks at the extreme locations in the four directions of the object (leftmost, rightmost, top, and bottom pixels). Polygon-RNN [21] treats the annotation problem as a polygon prediction task rather than a pixel prediction task. Polygon-RNN++ [22] upgrades its model based on Polygon-RNN and utilizes reinforcement learning to train the model. ScribbleSeg [47] is similar to click-based methods, where scribbles are converted into images and inputted into the model. With the emergence of large-scale datasets and continuous improvements in hardware capabilities, a universal large-model annotation algorithm called SAM [26] has appeared. SAM is capable of annotating objects in images using three methods: clicks, bounding boxes, and text. It also evaluates the quality of the output masks. SegNext [48] simultaneously converts three types of visual inputs (clicks, scribbles, and bounding boxes) into images and inputs them into the model.

B. DETR-Like Network Architecture

DETR [38] is a transformer-based framework that has achieved remarkable results in object detection. Due to its versatility and compatibility with diverse information sources, the DETR architecture has inspired models across various deep learning fields. Since the IIS discussed is a pixel prediction task akin to image segmentation, this section reviews the development of DETR-like models in segmentation.

MaskFormer [49] is the first to apply the DETR architecture to segmentation. It uses a pixel decoder to predict masks, while the transformer decoder captures global features through queries to achieve semantic segmentation, instance segmentation, or panoptic segmentation tasks. Mask2Former [50] is an improvement on MaskFormer. Mask2Former utilizes a deformable transformer [51] in the pixel decoder and proposes a mask attention in the transformer decoder to reduce computational complexity and memory usage. Mask DINO [52] combines detection and segmentation tasks, enhancing the accuracy of both. This showcases the extensibility of the DETR-based models. OneFormer [53] integrates a text encoder to jointly train multiple segmentation tasks, further demonstrating the extensibility of the DETR framework and its ability to handle multisource information.

C. Prompt Learning

Prompts in visual prompt learning can be categorized into three types: text prompts, visual prompts, and optimization prompts. A prompt is a set of feature vectors that guide the model in task-specific learning.

Text prompts are a method of using text to guide a model in completing visual tasks. CILP [54], [55], ALIGN [55] leverages contrastive learning to train a text–image multimodal encoder, aiming to obtain a pretrained feature extraction network. In this setup, text is designed as prompts through a

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text encoder and paired with images. In the field of image generation [56], prompts are similarly utilized in text format to guide the generation of images.

Visual prompts use images or image features to guide models in performing visual tasks. In SAM [26], click prompts and box prompts are formulated as the sum of category and position encodings as sparse prompts, and mask prompts are designed as a mask feature as dense prompts. Also, in Section II-A, most methods design click [17], [35], box [48], and scribble [47] prompts as a map, input into the model as dense prompts. In Piclick [32], click prompts are designed similar to SAM, with the addition of scale encodings.

Optimization prompts, often lacking direct practical significance, guide model learning by adjusting training objectives or loss functions. RSPrompter [4] trains a prompt generator to replace SAM's prompt generator for guiding remote sensing object detection tasks. Methods like VPT [57], HQ-SAM [58], and others [59], [60] add optimization prompts to pretrained models to adapt them for downstream tasks or enhance original task performance.

This article redesigns the DETR-like structure for IIS to improve annotation accuracy. PiClick, AdaptiveClick, and DETR-based segmentation methods are closely related, with PiClick and AdaptiveClick being interactive segmentation algorithms using DETR-like structures. However, in the transformer decoder, both PiClick and AdaptiveClick use attention mechanisms to emphasize annotated click locations. PiClick employs a learnable click prompt, while AdaptiveClick uses click map encoding. However, these methods primarily enhance click position information in global feature learning and provide limited benefit to mask feature generation. In contrast, our method enhances mask feature accuracy by redesigning the pixel decoder to emphasize prompts at annotated click positions, resulting in more precise mask predictions. In addition, queries in semantic segmentation aim to represent multiple categories and instance positions, necessitating large query numbers, such as 100 in Mask2Former and 300 in Mask DINO. However, IIS requires only foreground category prediction and does not need object position learning. Therefore, the transformer decoder in this article is designed to be more concise. Thus, our transformer decoder is streamlined by replacing category prediction with IoU prediction, enhancing global information learning and mask quality assessment.

III. Methods

The proposed model architecture, as shown in Fig. 3, comprises three main stages: multilevel feature extraction, click prompt and image interaction, and annotation mask output and quality evaluation. Section III-A details the overall network structure and multilevel feature extraction. Section III-B explains click prompt and image interaction, focusing on click prompt design and click prompt pixel decoder (CPPD). Section III-C covers annotation mask output and quality evaluation, including mask query initialization and lightweight transformer decoder (LTD). Section III-D describes the loss functions.

A. Network Structure

1) Multilevel Feature Extraction: As shown in Fig. 3(a), the input to the model head consists of the image $\mathcal{I} \in \mathbb{R}^{3\times H\times W}$, the positive and negative click maps $\mathcal{M}_{\text{clk.}} \in \mathbb{R}^{2\times H\times W}$ generated from the annotated clicks set $C = \{(x_1, y_1, l_1), \ldots, (x_N, y_N, l_N)\}$, and the previous segmentation mask $\mathcal{M}_{\text{prev.}} \in \mathbb{R}^{1\times H\times W}$, where $l_i \in \{0, 1\}, i \in \{0, \ldots, N\}$, 0 represents the negative click, 1 represents the positive click, and N represents the number of annotated clicks. $\mathcal{M}_{\text{prev.}}$ represents the previous segmentation result, which aids in stabilizing the model's predictions during the interactive process [17]. If no previous segmentation is available, $\mathcal{M}_{\text{prev.}}$ is a zero matrix. Typically, $\mathcal{M}_{\text{prev.}}$ is combined with $\mathcal{M}_{\text{clk.}}$ to form the interactive content map $\mathcal{M} \in \mathbb{R}^{3\times H\times W}$.

In order to enable sufficient information interaction between the clicks and the image within the backbone network, $\mathcal{I}$ and $\mathcal{M}$ undergo two separate patch embedding layers in the model head, then their outputs are summed. The resulting features are fed into the backbone to obtain multilevel features. Patch embedding layers convert the image into a series of patch embeddings. Each patch has a size of (P, P), and the dimensionality of the transformed features is D. Consequently, the final feature output from the head is denoted as $\overline{\mathcal{F}}_{\text{BB}} \in \mathbb{R}^{D\times HW/P^2}$, where BB means BackBone, and the overline indicates flattened 1-D features.

The backbone consists of ViT [44] and SimpleFPN [61]. ViT possesses powerful feature extraction capabilities and performs better than other networks in IIS [34]. However, the features output by ViT are single scale, which is not ideal for handling the large-scale variations present in remote sensing images. Therefore, we use the structurally simple and effective SimpleFPN to decompose these features into four scales. The final collection of multiscale features is $\overline{F} = \{\overline{\mathcal{F}}_1, \overline{\mathcal{F}}_2, \overline{\mathcal{F}}_3, \overline{\mathcal{F}}_4\}$, where $\overline{\mathcal{F}}_n \in \mathbb{R}^{D\times HW/(2^{(n-3)}P)^2}$, after unflattening $\overline{\mathcal{F}}_n$, $\mathcal{F}_n \in \mathbb{R}^{D\times(2^{3-n}H/P)\times(2^{3-n}W/P)}$. In other words, features $\mathcal{F}'_{\text{bb}} \in \mathbb{R}^{D\times(H/P)\times(W/P)}$, obtained after the backbone, are scaled by factors of 1/2, 1, 2, and 4 relative to the original side length.

2) Click Prompt and Image Interaction: After obtaining multiscale features $\overline{F}$, the three smallest sized features $\mathcal{F}_2, \mathcal{F}_3$, and $\mathcal{F}_4$ are flattened and concatenated to obtain $\overline{\mathcal{F}}^{\text{PD}}_{\text{attn.}}$, where $\overline{\mathcal{F}}^{\text{PD}}_{\text{attn.}} \in \mathbb{R}^{D\times\sum_{n=2}^{4}(HW/(2^{(n-3)}P)^2)}$ and PD means pixels decoder. In addition, multiscale click prompts $F_{\text{clk.}}$ are obtained from $\mathcal{F}_2, \mathcal{F}_3$, and $\mathcal{F}_4$, and mask features are obtained in the pixel decoder. After concatenating the click prompts form $F_{\text{clk.}}$, they are combined with $\overline{\mathcal{F}}^{\text{PD}}_{\text{attn.}}$ into the CPPD Layers, allowing the features at the clicks to interact with the image features. The specific details will be elaborated in Section III-B.

3) Annotation Mask Output and Quality Evaluation: The features obtained from the click prompt and image interaction part are at the three smallest scales, denoted as $\mathcal{F}'_2, \mathcal{F}'_3$, and $\mathcal{F}'_4$. To obtain the mask feature $\mathcal{F}_{\text{M}}$, $\mathcal{F}'_2$ is upsampled by a factor of 2, then summed with $\mathcal{F}_1$ to get $\mathcal{F}'_1$. In methods without transformer decoder [17], [62], [63], after obtaining the mask feature, the final segmentation result is often obtained directly through a 1×1 convolution. In methods with transformer decoder [32], [38], [50], global information from

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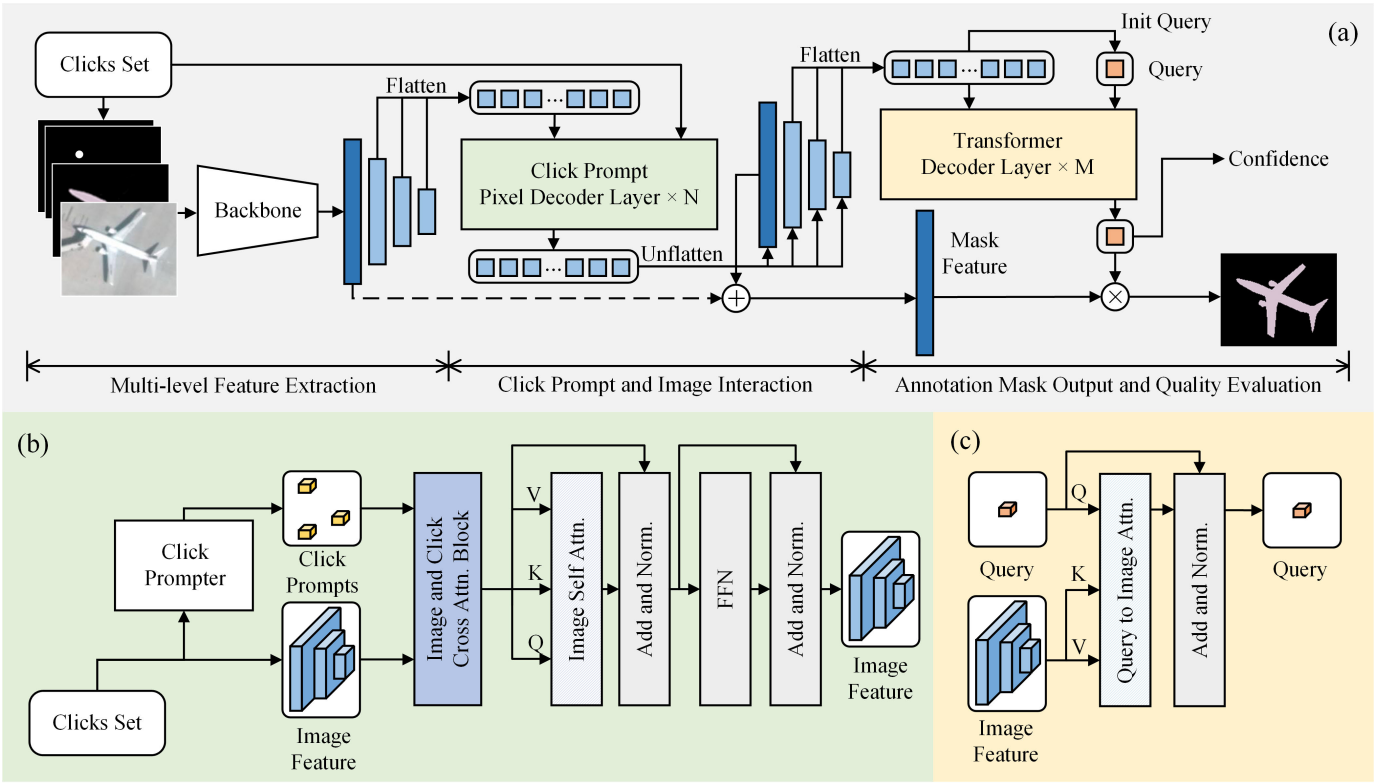


Fig. 3. Overview. (a) Overview of the entire network structure. (b) CPPD layer. (c) Transformer decoder layer.

the features is obtained using a mask query (or mask token) $\mathcal{F}_{\mathrm{MQ}} \in \mathbb{R}^{D \times N_q}$. $\mathcal{F}_{\mathrm{MQ}}$ is then multiplied with $\mathcal{F}_{\mathrm{M}}$ to obtain the final segmentation result. At the same time, $\mathcal{F}_{\mathrm{MQ}}$ will also pass through a feedforward neural network (FFN) to obtain confidence evaluations for segmentation masks. The specific details of our method will be elaborated in Section III-C.

B. Click Prompt Design and CPPD

In segmentation tasks, the pixel decoder generates mask features [49], [50], which are then used to obtain the segmentation results. Current IIS methods typically incorporate the click prompt into the transformer decoder [32], [37]. In contrast, the click prompt of our method is input into the pixel decoder, allowing each pixel to focus on the click prompts, which we believe is more effective for IIS tasks. This decoder is comprised of two main components: click prompt design and CPPD layer.

Based on SAM [26], DETR [38], and DINO [64], fully learnable parameters are used to represent information, such as click or bounding box data. Fully learnable parameters can slow model convergence. However, effective initialization can accelerate training and improve prediction accuracy. Thus, unlike SAM and PiClick, this work avoids representing click information through the summation of positional encoding and a learnable label embedding. The network inputs $\mathcal{M}_{\mathrm{clk.}}$ at the head of the backbone network, so the extracted features already contain label information of clicks. At the same time, since the fusion features with $\mathcal{M}_{\mathrm{clk.}}$ and $\mathcal{I}$ subjected to self-attention in the backbone network, the features obtained at each pixel position also contain semantic information.

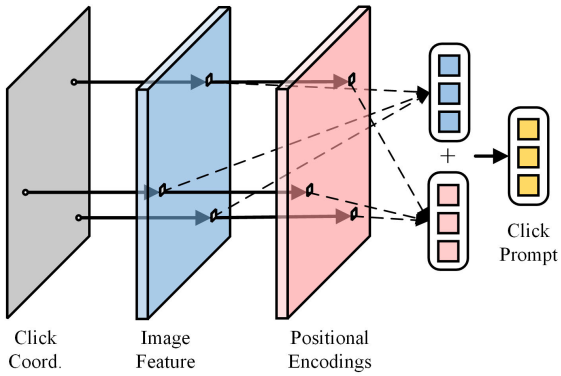


Fig. 4. Click prompt design. Gray represents the image. The circle in gray represents the click position. Blue represents the image feature. Pink represents positional encoding. Yellow represents the click prompt.

Therefore, our click prompt method directly obtains the feature of the annotation clicks at the positions corresponding to $\mathcal{F} \in \{\mathcal{F}_2, \mathcal{F}_3, \mathcal{F}_4\}$ and sums it with the positional encoding. In this way, the click prompt simultaneously incorporates positional information, label information, and semantic information, as shown in Fig. 4. The feature acquisition method on a single level is illustrated as follows:

$$\begin{aligned} \mathcal{F}_{\mathrm{clk.},n} &= \sigma(\mathcal{F}, \mathbf{c}) + \sigma(\mathrm{PE}, \mathbf{c}) \\ \mathbf{c} &= (i, j), \quad \mathcal{F} \in \{\mathcal{F}_2, \mathcal{F}_3, \mathcal{F}_4\} \end{aligned} \tag{1}$$

where n represents the nth click, (i, j) represent matrix coordinates of the annotated click. $\sigma(\mathcal{F}, \mathbf{c})$ means that the coordinate $\mathbf{c}$ sampled on $\mathcal{F}$ and sampling method is bilinear interpolation.

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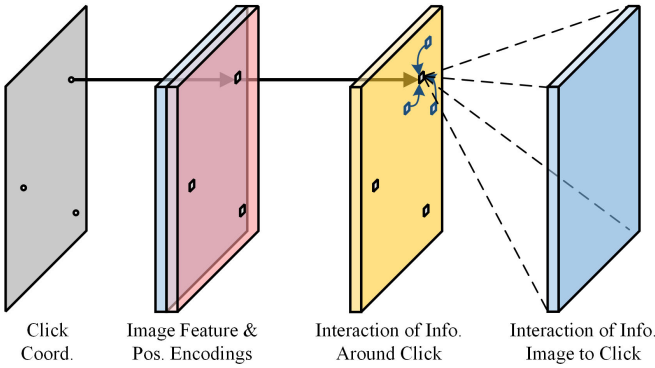


Fig. 5. ICCAB, using a click as an example. Gray represents the image. The circle in gray represents the click position. Blue represents the image feature. Pink represents positional encoding. Yellow represents the click prompt, but only the features at and around the selected click positions need to be computed.

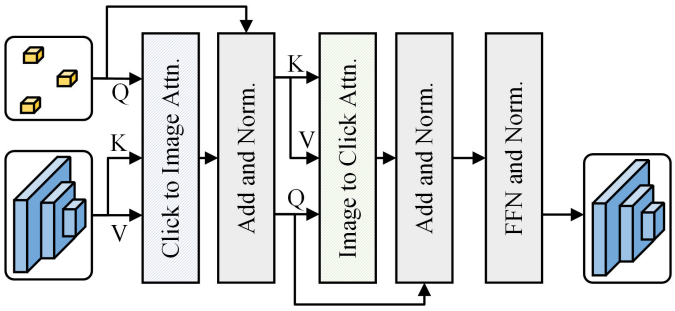


Fig. 6. ICCAB structural diagram. Click-to-image attention uses deformable attention, and image-to-click uses multihead attention.

PE represents the position encoding, and the size is the same as $\mathcal{F}$. After sampling features for N clicks at all three scales, we obtain $F_{\text{clk.}} = \{\mathcal{F}_{\text{clk.},n,l} | n = 1, \ldots, N, l = 2, 3, 4\}$. After concatenation, the representation of clicks is $\mathcal{F}_{\text{clk.}} \in \mathbb{R}^{D\times 3N}$.

After obtaining click prompts with rich information representation, we designed a composite attention to enable full interaction between click prompts and image, called image and click cross attention block (ICCAB) as in Fig. 6. We first employ deformable attention [51] to allow $\mathcal{F}_{\text{clk.}}$ at click positions to interact with surrounding pixels and obtain a comprehensive click representation $\mathcal{F}'_{\text{clk.}}$. Then, through $\mathcal{F}'_{\text{clk.}}$, we apply multihead attention [65] with the features of the entire image, enabling the information at each click position to influence the image features, as shown in Fig. 5.

The equation for the click-to-image deformable attention is shown in the following equation as a simple representation:

$$\begin{aligned} \mathcal{F}'_{\text{clk.}} &= A_{qk}\sigma(\mathcal{F}_{\text{clk.}}, \mathbf{c} + \Delta\mathbf{c}), \\ A_{qk} &= W_1\mathbf{z}_q + b_1, \quad \Delta\mathbf{c} = W_2\mathbf{z}_q + b_2, \ \mathbf{z}_q = \sigma(\mathcal{F}_{\text{clk.}}, \mathbf{c}) \end{aligned} \tag{2}$$

where $\mathcal{F}_{\text{clk.}} \in \{\mathcal{F}_{\text{clk.},1}, \ldots, \mathcal{F}_{\text{clk.},N}\}$, and $\mathbf{c} \in C'$, $C' = \{(i_1, j_1), \ldots, (i_N, j_N)\}$. Based on the coordinates of the annotated clicks $\mathbf{c}$, features $\mathbf{z}_q$ are extracted from $\mathcal{F}_{\text{clk.}}$ and treated as a query in the attention structure. Through a linear layer, attention weights A_{qk} and coordinate biases $\Delta\mathbf{c}$ are learned from the query. Value in the attention structure extracts features from the clicks with added biases, as $\sigma(\mathcal{F}, \mathbf{c}+\Delta\mathbf{c})$ in (2), focusing the output features $\mathcal{F}'_{\text{clk.}} \in \mathbb{R}^{D\times 1}$ on the positions surrounding $\mathbf{c}$.

The equation for the image-to-click multihead attention is shown in the following equation as a simple representation:

$$\mathcal{F}' = \mathcal{F}\mathcal{F}'^{\mathsf{T}}_{\text{clk.}}\mathcal{F}'_{\text{clk.}}. \tag{3}$$

This is a standard multihead attention structure where image features $\mathcal{F}$ act as queries, and $\mathcal{F}'_{\text{clk.}}$ act as keys and values, thereby modifying the image feature. $\mathcal{F}'^{\mathsf{T}}_{\text{clk.}}$ represents the transpose of $\mathcal{F}'_{\text{clk.}}$. It is important to note that here, unlike conventional methods, where $\mathcal{F}_{\text{clk.}}$ is treated as query, we treat $\mathcal{F}$ as query. We aim to enhance the representation of click information in the mask feature.

To simplify the description and remove unnecessary details, the above-described deformable attention and multihead attention are both simplified representations. The details involving multiple heads, multiple scales, attention matrix scale, and skip connections are described in detail in [51] and [65].

As shown in Fig. 3(b), after obtaining the modified features from ICCABs, the feature undergoes an image self-attention and an FFN. This feature is the final output feature of the pixel decoder layer. The self-attention operation uses deformable attention. The pixel decoder layer is a recurrent structure. Given inputs $\mathcal{F}_2$, $\mathcal{F}_3$, and $\mathcal{F}_4$, and C', it continuously updates $\mathcal{F}_2$, $\mathcal{F}_3$, and $\mathcal{F}_4$ until outputting $\mathcal{F}'_2$, $\mathcal{F}'_3$, and $\mathcal{F}'_4$.

C. Mask Query Initialization and LTD

In DETR-like models used for object detection [38] and semantic/instance segmentation [49], [50], the transformer decoder typically utilizes queries to perform attention operations on the image, extracting global information for predictions. These tasks generally output object boxes, classes, and mask information. However, in IIS, segmentation typically involves a single category, representing foreground objects. While some approaches adopt the DETR-like model and Hungarian matching training method, separating segmentation into foreground and background [32], [37], we find this unnecessary. Direct foreground segmentation avoids the computational overhead of Hungarian matching and solves query homogenization during matching. Furthermore, previous DETR-like models for IIS placed click prompts in the transformer decoder. To better integrate click information into the mask feature, we have placed the click prompts in the pixel decoder, allowing the transformer decoder to focus on quickly learning a global representation for accurate segmentation.

We used deformable attention between the query and mask feature in the LTD layer. Deformable attention has fewer parameters and computational requirements and can get global information by focusing on localized regions. To achieve a more accurate global representation, the mask query predicts the IoU of the mask. In addition, we improved the initialization of the mask query to facilitate learning the semantic details of the annotated objects, aiming to accelerate model convergence.

Mask query initialization involves directly appending an IoU prediction head from $\overline{\mathcal{F}}'_2$, $\overline{\mathcal{F}}'_3$, and $\overline{\mathcal{F}}'_4$. The parameters of the IoU prediction head are shared with the final IoU prediction head for mask query. We extract the most confident feature

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TABLE I
COMPLEXITY OF ATTENTION MODULE IN TRANSFORMER DECODER

Module	Complexity
Deformable Attention	$O_{\mathrm{DA}} = 2N_qC^2 + \min\left(HWC^2, N_qKC^2\right)$
Mask Attention	$O_{\mathrm{MA}} = N_qC^2 + N_mC^2 + N_qN_mC$
Multi-head Attention	$O_{\mathrm{A}} = N_qC^2 + HWC^2 + N_qHWC$

from the image features to initialize the mask query. The formula for obtaining the initialized mask query is shown as follows:

$$\begin{aligned}\mathbf{c}_{\mathrm{MQ}}, l &= \operatorname{argmax}\left(\operatorname{EvalHead}\left(*\overline{\mathcal{F}}'_2, *\overline{\mathcal{F}}'_3, *\overline{\mathcal{F}}'_4\right)\right)\\ \mathcal{F}_{\mathrm{MQ}} &= \sigma\left(\mathcal{F}'_l, \mathbf{c}_{\mathrm{MQ}}\right)\end{aligned} \tag{4}$$

where argmax(·) denotes obtaining the coordinates $\mathbf{c}$ and level l of the feature where the maximum confidence feature is located. EvalHead(·) is an FFN head to evaluate the confidence of the feature. $\mathcal{F}_{\mathrm{MQ}}$ is mask query, $\mathcal{F}_{\mathrm{MQ}} \in \mathbb{R}^{D\times N_q}$, $N_q = 1$, and $*$ denotes splitting the feature matrix along the HW dimension.

After obtaining the $\mathcal{F}_{\mathrm{MQ}}$, we perform deformable attention with mask query to the image. The formula is represented as follows:

$$\begin{aligned}\mathcal{F}'_{\mathrm{MQ}} &= A_{qk}\sigma\left(\mathcal{F}', \mathbf{c}_{\mathrm{MQ}} + \Delta\mathbf{c}_{\mathrm{MQ}}\right)\\ A_{qk} &= W_1\mathcal{F}_{\mathrm{MQ}} + b_1, \quad \Delta\mathbf{c}_{\mathrm{MQ}} = W_2\mathcal{F}_{\mathrm{MQ}} + b_2.\end{aligned} \tag{5}$$

Here, we also utilize deformable attention to attend to $\mathcal{F}' \in \{\mathcal{F}'_2, \mathcal{F}'_3, \mathcal{F}'_4\}$. The attention involves clicks around $\mathbf{c}_{\mathrm{MQ}}$ to update $\mathcal{F}_{\mathrm{MQ}}$. In addition, because deformable attention only performs attention computations with points near the reference points, it is faster and more lightweight.

As shown in Fig. 3(c), there are skip connections and regularization both before and after LTD. This forms a recurrent structure where $\mathcal{F}_{\mathrm{MQ}}$ is updated multiple times.

After obtaining the final $\mathcal{F}'_{\mathrm{MQ}}$ and mask feature, we use (6) and (7) to obtain the confidence and the final predicted mask:

$$\mathcal{M}_{\mathrm{pred.}} = \operatorname{MaskHead}\left(\mathcal{F}'_{\mathrm{MQ}} \cdot \mathcal{F}_{\mathrm{M}}\right) \tag{6}$$

$$\mathcal{S}_{\mathrm{conf.}} = \operatorname{IoUHead}\left(\mathcal{F}'_{\mathrm{MQ}}\right) \tag{7}$$

where MaskHead(·) and IoUHead(·) are the prediction heads composed of linear layers and $\mathcal{S}_{\mathrm{conf.}}$ means the predicted IoU.

In the current DETR-like models for IIS [32], [37], the transformer decoder primarily utilizes mask attention, similar to Mask2Former. Also, multihead attention is a common module in the transformer decoder of DETR-like model [38], [49]. The complexity of deformer attention, mask attention, and multihead attention modules is shown in Table I, and are denoted by O_{DA}, O_{MA}, and O_{A}. In O_{DA}, $N_q = 1$, K is the number of sampling points for three scales, and $K = 4 \times 3$. HW for three scales can be denoted as $21/16N_{\mathcal{F}'_2}$, where $N_{\mathcal{F}'_2}$ is the number of pixels in $\mathcal{F}'_2$. In practice, $21/16N_{\mathcal{F}'_2} \gg N_qK$, so $O_{DA} = N_qC^2 + N_q(K+1)C^2$. In O_{MA}, N_m is the number of mask pixels for three scales. $21/16N_{\mathcal{F}'_2} > N_m \gg N_q(K+1)$ in practice, so $O_{\mathrm{A}} > O_{\mathrm{MA}} > O_{\mathrm{DA}}$. So, deformable attention has the lowest complexity among these attentions. In our transformer decoder, the EvalHead(·) is a linear layer, and the complexity is $N_{\mathcal{F}'_2}C^2$. The entire transformer decoder only requires one EvalHead(·). In the transformer decoder of Mask2Former, each mask requires a linear layer for output, and the complexity is $21/16N_lN_{\mathcal{F}'_2}C^2$, where N_l is the number of layers. Despite the transformer decoder relying on multihead attention without additional structures, its inherent complexity remains significant. Also, the linear layers interspersed between MaskHead(·), IoUHead(·) (which are classification heads in other models, with the same complexity), and attention structures are common modules shared by all models. Therefore, our transformer decoder has lower complexity during the inference process when having the same N_l. Furthermore, compared with other methods, our approach ($N_q = 1$) allows for direct training without the need for Hungarian matching. This significantly reduces the computational complexity of the model during training. Overall, our transformer decoder is lighter.

D. Loss Function

Since this network uses a single query, the training process does not require Hungarian matching like DETR-like models. A simple compound loss is sufficient.

The loss for the IoU prediction function is shown in (8), while the loss for the mask is calculated using the normalized focal loss (NFL) as shown in (9)

$$\mathcal{L}_{\mathrm{conf.}} = \left\| \frac{\left(\mathcal{M}_{\mathrm{pred.}} > \mathrm{threshold}\right) \cup \mathcal{M}_{\mathrm{GT}}}{\left(\mathcal{M}_{\mathrm{pred.}} > \mathrm{threshold}\right) \cap \mathcal{M}_{\mathrm{GT}}} - \mathcal{S}_{\mathrm{conf.}} \right\| \tag{8}$$

$$\begin{aligned}\mathcal{L}_{\mathrm{mask}} &= \sum_{i,j}^{N} \frac{-\left(1-p_{i,j}\right)^{\gamma}\log\left(p_{i,j}\right)}{\sum_{i,j}^{N}\left(1-p_{i,j}\right)^{\gamma}}\\ p_{i,j} &= \left(\mathcal{M}_{\mathrm{pred.}} \uparrow\right)_{i,j}\end{aligned} \tag{9}$$

where $\mathcal{M}_{\mathrm{GT}}$ is the ground truth mask, threshold is the threshold for the mask, γ is the focal loss parameter, $\mathcal{M}_{\mathrm{pred.}} \uparrow_{i,j}$ is the predicted mask value at position (i, j), and $\uparrow$ denotes upsampling to the original image size.

The final loss is $\mathcal{L} = \alpha\mathcal{L}_{\mathrm{conf.}} + \beta\mathcal{L}_{\mathrm{mask}}$.

IV. EXPERIMENTS

A. Datasets

We chose the high-resolution scale (0.05–15 m) as the primary training data because there are numerous categories at this scale. Following the common practice in the IIS task [17], [34], [35], we categorized these classes into instance categories (or called thing categories) and noninstance categories (or called stuff categories). We selected six datasets to serve as the training and testing data, respectively.

iSAID [66] is an instance segmentation dataset sourced from multiple sensors and platforms, primarily comprising optical images (RGB three bands) with a few panchromatic images. It ranges in resolution from 0.5 to 5 m and encompasses 15 categories across 2806 images, with image widths ranging from 800 to 13 000 pixels. The categories are all considered instance categories. This dataset covers most instance classes in high-resolution images.

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ISPRS Vaihingen[1] is a semantic segmentation dataset derived from aerial sensor imaging, with near-infrared, red, and green bands. It consists of 33 images at a resolution of 0.09 m, with image widths ranging from 1900 to 4000 pixels. There are six categories, with buildings, trees, and vehicles serving as instance categories, while impervious surfaces, low vegetation, and background are noninstance categories.

ISPRS Potsdam,[2] similar to ISPRS Vaihingen, is a semantic segmentation dataset sourced from aerial sensor imaging, with near-infrared and RGB bands. It comprises 38 images at a resolution of 0.05 m, all with a width of 6000 pixels, and the same categories as Vaihingen. Both Vaihingen and Potsdam are commonly used in remote sensing research, covering common urban scene categories.

WHU Building [67] is a dataset specifically tailored for building-related applications in remote sensing images. This study uses two datasets for semantic segmentation: Satellite Dataset I and Satellite Dataset II. Satellite Dataset I includes images from various cities worldwide, captured by satellites such as QuickBird, Worldview series, IKONOS, and ZY-3, with visible light bands and 204 images at resolutions ranging from 0.3 to 2.5 m. Satellite Dataset II focuses on East Asia, comprising adjacent images from the same satellite. It contains 17 388 images at a resolution of 0.45 m, all with a width of 512 pixels, and 25 749 buildings. The building is considered an instance category.

NWPU [68] is an object detection dataset that later introduced an instance segmentation version [69], sourced from aerial sensors, primarily comprising optical image (RGB three bands) with a few panchromatic images. It ranges in resolution from 0.08 to 2 m across ten categories. This study uses the positive image set, consisting of 650 images with widths ranging from 500 to 2000 pixels. The categories are all considered as instance categories.

SSDD [70] is a dataset for ship instance segmentation captured using SAR imaging, with resolutions ranging from 1 to 15 m across ten images, containing a total of 1160 instances. The ship is considered an instance category.

To maintain consistency with the IIS task in natural images and to enhance annotation accuracy and model scalability, the objects considered as instance categories in the semantic segmentation datasets were converted into panoptic segmentation datasets using connectivity analysis in OpenCV, as shown in Fig. 7. The training dataset includes iSAID, ISPRS Potsdam, and Satellite Dataset II from WHUBuilding, covering most instance objects in high-resolution images. During training, images from iSAID and ISPRS Potsdam were cropped to a size of 800×800 pixels, and patches without masks were removed. The testing dataset comprises ISPRS Vaihingen, Satellite Dataset I from WHUBuilding, NWPU, and SSDD. Testing images vary in imaging conditions, environments, resolutions, and imaging sensors compared with the training data. In addition, RGB bands were used for training ISPRS Potsdam.

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

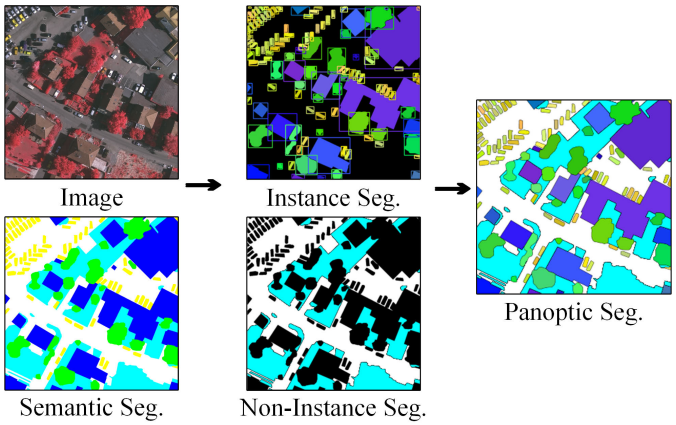


Fig. 7. For a given image and the ground truth of semantic segmentation. We convert instance categories (buildings, trees, and vehicles) into instance segmentation data, while keeping noninstance categories (impervious surfaces and low vegetation) unchanged. Merge the two to obtain a panoptic segmentation data.

B. Implementation Details

1) Multilevel Feature Extraction Part: We use ViT-base (ViT-B) and ViT-small (ViT-S) as our feature encoding networks. The patch embedding size for both networks is $(16, 16)$, which means $P = 16$. ViT-B is pretrained by MAE [71] on ImageNet-1K [72], with an output feature dimension of 768. ViT-S is pretrained as DINO [73] on ImageNet-1K, with an output feature dimension of 384. SimpleFPN converts the output single-scale features into multiscale features and then unifies the dimensions using 1×1 convolution. For ViT-B, the dimensions of multiscale features are (192, 384, 768, and 768) and for ViT-S they are (96, 192, 384, and 384). Afterward, all are uniformly adjusted to 256 feature dimensions.

2) Click Prompt and Image Interaction Part: In the click prompt, the position encoding is composed of sine and cosine functions [38]. The deformable click-to-image attention in ICCAB has eight heads, sampling three scales, with four points sampled for each scale. The image-to-click attention in ICCAB has eight heads. The image deformable attention has eight heads, sampling three scales, with four points sampled for each scale. The number of CPPD layers is 2.

3) Annotation Mask Output and Quality Evaluation Part: The deformable attention with mask query to image has eight heads, samples across three scales, and samples four points per scale. The number of transformer decoder layers is 3.

4) Setup of Training: Training data is divided based on the dataset's data volume, with proportions of 0.7, 0.15, and 0.15 for iSAID, ISPRS Potsdam, and Satellite Dataset II of WHU Building, respectively. The model is trained for 60 epochs with the Adam optimizer. The initial learning rate is 5×10^{-5} and reduces by ten times in the 50th epoch. In the loss function, $\alpha = 1$ and $\beta = 20$. The batch size is 16 for ViT-B and 8 for ViT-S. All models are trained and tested on Linux with an NVIDIA RTX 3090Ti GPU. In the training process, the initial click generation method is adopted similar to DOS [35]. In addition, similar to RITM [17], during the training process, annotated clicks are generated, with a

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TABLE II

COMPARISON IN NOC80, NOC85, AND NOC90 BETWEEN OUR METHOD AND STATE-OF-THE-ART METHODS TESTED ON NWPU, SSDD, SATELLITE DATASET I FROM WHUBUILDING (ABBREVIATED AS WUHB-I), AND ISPRS VAIHINGEN. THE BEST IS BOLDED AND THE RUNNER-UP IS UNDERLINED

Method	Backbone	NWPU			SSDD			WHUB-I			ISPRS Vaihingen		
		NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90
RITM	ResNet50	2.98	4.37	7.47	4.28	8.59	15.36	3.60	5.08	8.73	4.66	5.94	8.93
CDNet	ResNet50	5.16	7.68	11.26	9.85	15.24	19.30	6.04	9.11	14.37	7.38	9.96	13.93
GPCIS	ResNet50	4.28	6.01	9.37	7.00	11.74	17.23	4.72	6.84	11.52	6.30	8.04	11.31
RITM	HRNet32	2.86	4.13	7.06	3.94	7.84	14.70	3.46	4.84	8.17	4.33	5.48	8.09
SAM*	ViT-B	6.39	8.82	11.72	12.17	16.89	19.45	7.37	11.02	15.86	8.08	11.02	15.51
SAM*	ViT-H	5.70	8.02	11.17	10.09	15.70	19.16	7.70	11.16	15.84	7.43	10.08	14.35
SimpleClick	ViT-B	2.74	3.95	6.99	3.28	6.54	13.43	3.42	4.67	7.84	4.15	5.28	7.86
PiClick**	ViT-B	3.17	4.68	7.85	3.45	7.19	14.96	3.48	4.73	7.92	4.74	6.04	8.93
AdaptiveClick	ViT-B	3.02	4.48	7.52	4.60	8.90	15.04	3.28	4.63	8.01	4.49	5.76	8.68
Ours	ViT-S	2.85	4.12	6.99	**3.07**	**6.19**	**12.93**	3.44	4.66	7.66	4.50	5.70	8.25
	ViT-B	**2.52**	**3.63**	**6.49**	3.49	7.09	13.55	**3.25**	**4.38**	**7.20**	**3.87**	**4.96**	**7.39**

* refers to SAM directly adopting the pre-trained parameters without retraining them.

** indicates that PiClick did not employ multi-output prediction and training, with both the model and the loss function remaining the same as PiClick's.

maximum loop count of 3. During the training process, the input image size was 448×448.

C. Evaluation Metrics

Consistent with previous work [17], [34], [35], we use the average number of clicks (NoC) required to achieve a certain IoU as the primary evaluation metric. The maximum number of annotated clicks is set to 20. When the number of annotated clicks reaches 20, the annotation is considered failed, and the object is recorded as annotated with 20 clicks. We set the IoU thresholds for the objects as 0.80, 0.85, and 0.90, respectively. NoC80 refers to the average NoC required to achieve an IoU of 0.80. During the testing process, clicks are automatically generated by the algorithm, and the click positions are marked at the center of the region with the largest annotation error.

D. Comparative Experiment

We compared our method with seven state-of-the-art methods. RITM [17], GPCIS [74], and CDNet [33] are network architectures based on CNN. SAM [26], SimpleClick [34], PiClick [32], and AdaptiveClick [37] are network architectures based on ViT. Among them, PiClick and AdaptiveClick are DETR-like networks.

SAM is a prompt-based segmentation model trained on a large-scale dataset. Due to the unavailability of its training dataset and the complexity of replicating its training process, we utilized its pretrained parameters for testing. PiClick requires a parent–child relationship among dataset categories to generate multiannotation outputs. However, since public remote sensing datasets lack such category relationships, we trained PiClick using a version that does not output multiple results. All models, except SAM, were retrained on the combined remote sensing dataset using consistent datasets and image augmentation methods, with hyperparameters set to default values from the open-source code.

1) Quantitative Comparison: The comparison results of state-of-the-art methods are shown in Table II. Overall, our method achieved the greatest advantage in NoC80, NoC85, and NoC90.

Regarding the size of the backbone networks, ViT-S has slightly fewer parameters (22M) than ResNet-50 (23M) and significantly fewer than HRNet-32 (31M). ViT-B has 86M parameters, and ViT-H has 631M parameters.

At the ViT-S parameter scale, our method outperforms all the models in the comparison. This suggests our method is particularly effective for high-quality annotations. On the SSDD dataset, our method outperforms even many ViT-B-based algorithms, showing strong cross-domain performance.

At the ViT-B parameter scale, our approach demonstrated notable advantages across all datasets except SSDD, which is an SAR dataset, whereas our training data focused on optical imagery. As model parameters increase, overfitting to training data becomes more likely, potentially leading to diminished performance on varied domains. Despite this, our method remains highly competitive, and on other datasets, especially in NoC90, it consistently outperformed existing methods, highlighting its strength.

Fig. 8 shows convergence analysis of various methods on four testing datasets. It can be observed that, except for SSDD, our ViT-B-based methods demonstrate significant advantages after just two clicks, with around ten clicks typically achieving an annotation accuracy near 90%. On the SSDD dataset, despite domain differences, our ViT-B method remains strong among ViT-S methods. Overall, our method achieves higher IoU annotation masks and reduces the number of required interactions.

Fig. 9 presents a histogram analysis across the testing datasets. With only one click, most methods struggle to achieve a high IoU for many samples. In the IoU@1 of Fig. 9, only around 15% of samples achieve an IoU between 0.8 and 1.0 with a single click. After the second click, our method shows clear superiority, with many samples reaching IoU levels between 80% and 100%. Furthermore, our method consistently produces a greater proportion of high-IoU samples, offering higher quality annotations with fewer interactions.

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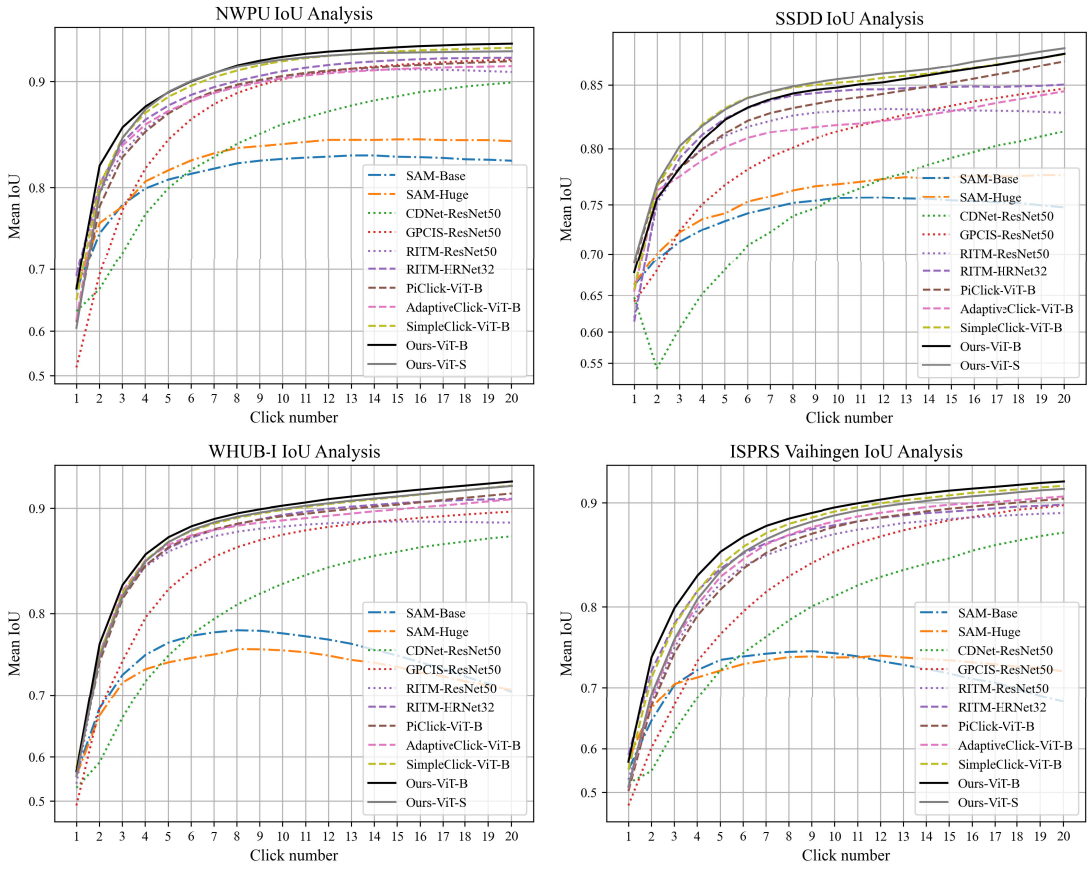


Fig. 8. Convergence analysis for NWPU, SSDD, WHUB-I, and ISPRS Vaihingen. The horizontal axis represents the NoC, and the vertical axis represents the mean IoU.

2) Complexity Comparison: We calculated the number of parameters, floating point operations (FLOPs), memory usage, time spent per interaction, and time spent evaluating the testing datasets for the state-of-the-art methods, as summarized in Table III. For FLOPs, memory usage, and time spent per interaction, the input image size is (448, 448), and the NoC is 20. Memory usage and time spent per interaction were averaged over 1000 tests. Time spent evaluating the testing datasets was measured under the same conditions used for quantitative comparison, with all tests conducted on a single RTX 3090.

Overall, our model's inference speed is moderate relative to other models, achieving real-time performance with the available hardware. Notably, compared with DETR-like methods, our approach significantly reduces runtime due to its redesigned pixel decoder and LTD. In addition, the ICCAB not only lessens the computational load but also enhances multiscale performance, making it particularly effective for remote sensing images.

3) Visualization Comparison: Comparing SAM with our method, it is evident that SAM, trained on natural images, lags behind models specifically tuned for remote sensing datasets in click-based annotation. SAM struggles with remote sensing image semantics and exhibits lower stability during the interactive process.

As shown in Fig. 10, SAM's stability is inferior to ours with multiple clicks. For instance, it struggles to correct partial shadows on airplanes (rows 1 and 2) and fails to address line noise in SAR images (fifth row and sixth column). The model even crashes in some instances (fifth row and seventh column). In addition, SAM does not correct building shadows (seventh row and seventh column) and generates annotation errors in various images (third row and seventh column).

We also provided a visual representation of the supervised methods, as shown in Fig. 11. The results show that our method excels in annotating small objects, such as the vehicles in the fourth row and the buildings in the ninth and tenth rows. While the IoU for the vehicles in the fourth row is slightly lower than PiClick's, our method still captures the vehicles' main shape effectively, with the discrepancy being minor.

Our method demonstrates superior edge and shape prediction compared with others. It also shows better performance in cross-domain annotations. For example, it effectively segments ships in SAR images (seventh and eighth rows), while other methods struggle with microwave noise.

Moreover, our method achieves more accurate annotations for noninstance class objects, as seen in the sixth row. By leveraging multiscale features and interaction with click locations in the pixel decoder, our approach provides precise annotation results for both large noninstance class objects and small objects, maintaining strong performance across general and cross-domain scenarios.

E. Ablation Study

1) Comparison of Layer Numbers: To validate the hyperparameter choices in our model, we conducted six comparative

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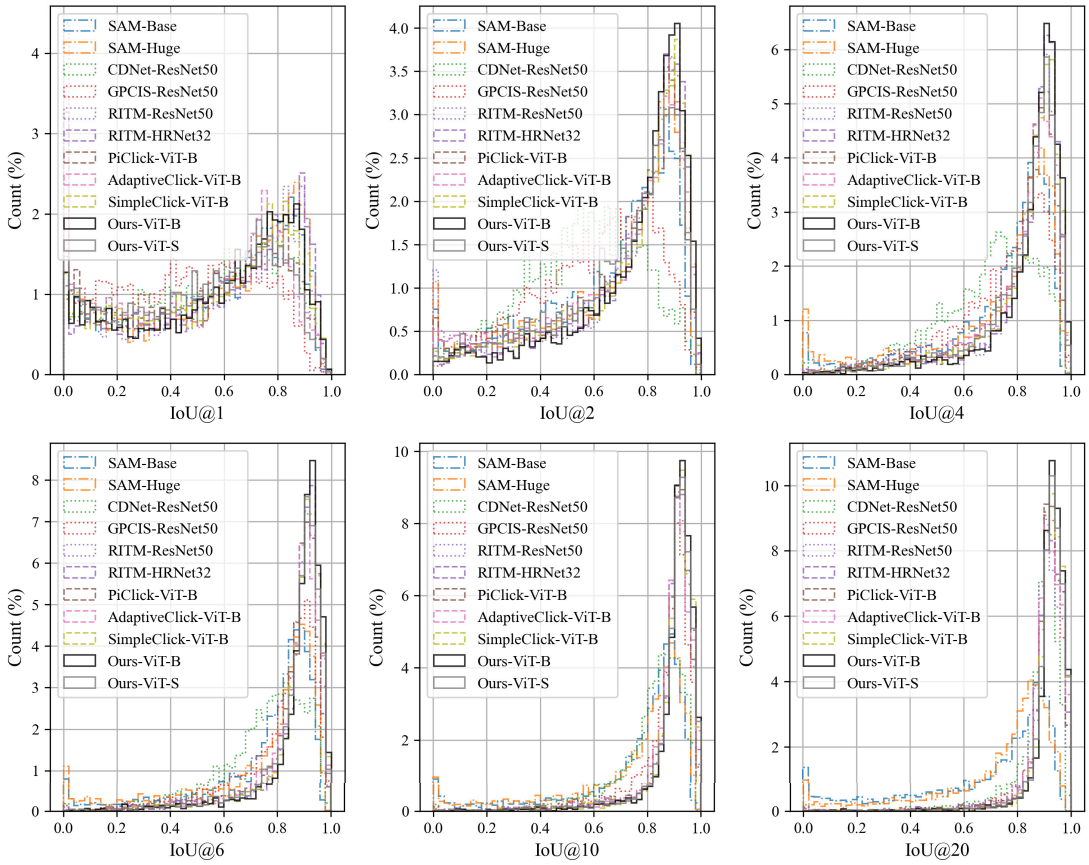


Fig. 9. Histogram analysis for combined datasets of NWPU, SSDD, WHUB-I, and ISPRS Vaihingen. The horizontal axis represents IoU, and the vertical axis represents the count of all objects. IoU@N represents the IoU with N clicks.

TABLE III

COMPUTATION COMPARISON FOR MODEL PARAMETERS, FLOPS, MEMORY, TIME SPENT PER INTERACTION, AND TIME SPENT EVALUATING THE TESTING DATASETS OF NWPU, SSDD, WHU-I, AND ISPRS VAIHINGEN

Method	Backbone	Parameters	FLOPs	Memory	Time Spent Per Interaction	Time Spent Evaluating (Hours:Minutes:Seconds)			
						NWPU	SSDD	WHUB-I	ISPRS Vaihingen
RITM	ResNet50	31.41M	213.87G	1.03G	16.63ms	0:24:33	0:23:29	1:05:58	0:21:04
CDNet	ResNet50	39.90M	270.30G	1.35G	18.65ms	0:36:49	0:32:53	1:48:52	0:34:20
GPCIS	ResNet50	39.39M	271.69G	1.97G	24.49ms	0:24:58	0:21:49	1:04:31	0:21:21
RITM	HRNet32	30.95M	103.74G	0.88G	41.11ms	0:36:40	0:42:37	1:50:02	0:31:38
SimpleClick	ViT-B	96.46M	169.78G	0.87G	18.30ms	0:24:16	0:23:03	0:59:53	0:19:48
PiClick	ViT-B	97.56M	197.40G	1.09G	54.08ms	0:50:40	0:57:54	2:10:43	0:44:07
AdaptiveClick	ViT-B	116.41M	269.81G	1.28G	47.89ms	0:44:45	0:52:49	2:01:33	0:40:28
Ours	ViT-S	26.89M	67.33G	0.96G	25.26ms	0:25:50	0:25:20	1:05:40	0:22:48
	ViT-B	93.50M	185.96G	1.40G	38.95ms	0:32:25	0:38:12	1:28:19	0:26:40

experiments using ViT-H as the backbone, as sdetailed in Table IV.

The results show optimal performance when the CPPD number is 2 and the LTD number is 3. Performance on the SSDD dataset varies significantly, likely due to the challenges in annotating SAR images, which are often noisy, of poor quality, and have fuzzy boundaries.

The performance of CPPD improves rapidly with an increasing number of layers. This is due to CPPD's composite structure, which includes deformable attention and multihead attention, providing strong information representation capabilities. However, stacking multiple CPPD layers tends to convert the model structure toward self-attention, weakening the impact of click prompts and resulting in blurred information at the click locations and reduced overall performance.

For the number of LTD, we aim to maintain the best performance while keeping the model lightweight. Therefore, we conducted experiments with three different quantities and selected the optimal result.

2) Multilevel Feature Extraction Ablation Experiment: We conducted ablation studies on the multilevel feature extraction component from two perspectives.

First, we compared performance between single-scale and multiscale features. Single-scale features were taken from the second-largest scale, $\mathcal{F}_2$. Backbone uses ViT-S+SimpleFPN.

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Fig. 10. Segmentation results of our method and SAM on NWPU (lines 1–4), SSDD (lines 5 and 6), WHUB-II (lines 7 and 8), and ISPRS Vaihingen (lines 9 and 10). The red masks are segmentation results, green points are positive clicks, and red points are negative clicks. Green boxes represent the zoomed-in-view area. The expected mask is segmentation from datasets.

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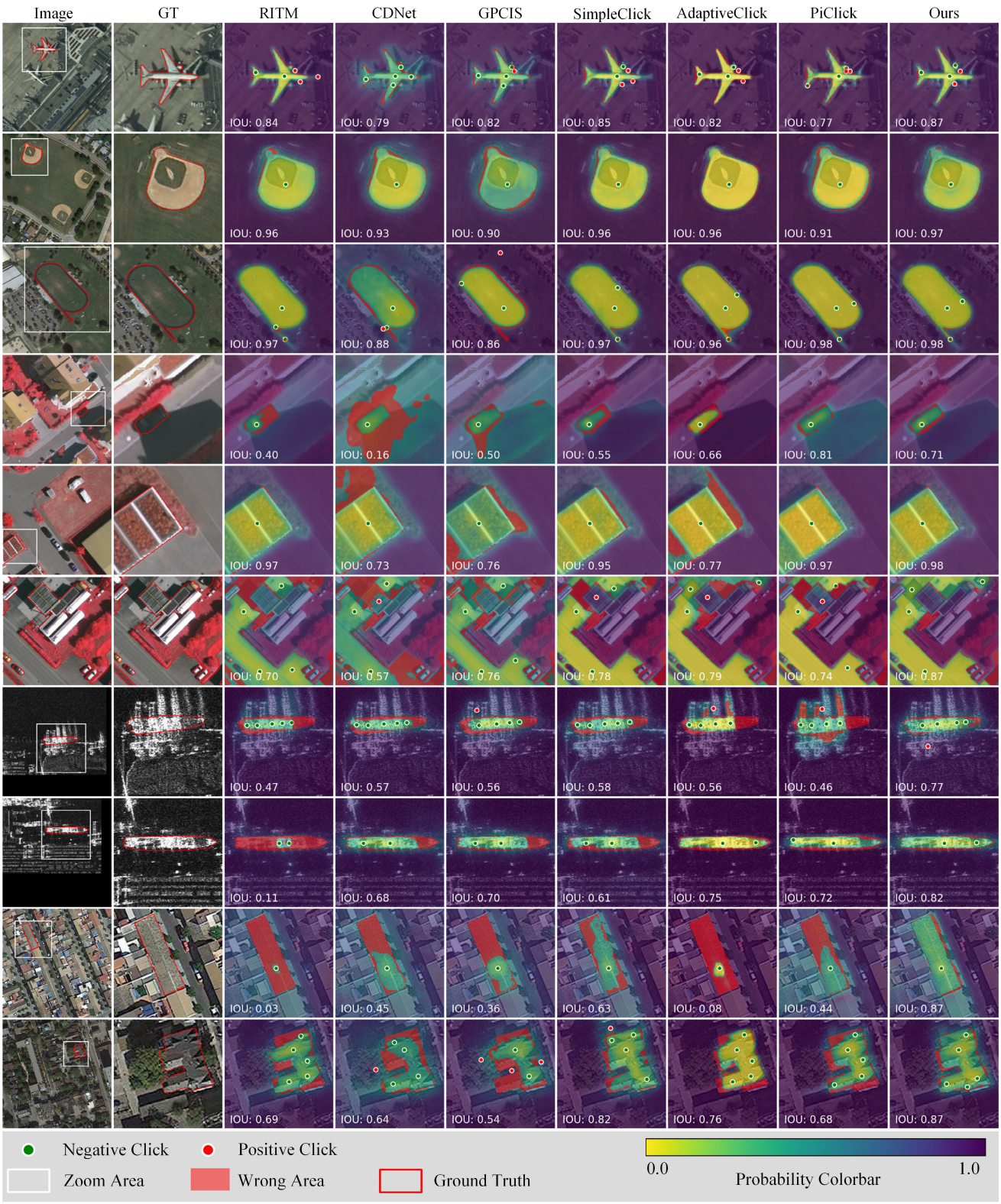


Fig. 11. Segmentation results of RITM (HRNet32), CDNet (ResNet50), GPCIS (ResNet50), SimpleClick (ViT-B), AdaptiveClick (ViT-B), PiClick (ViT-B), and Ours (ViT-B).

Results in Table V indicate that single-scale features are inferior to multiscale features for remote sensing images. Multiscale features offer more comprehensive scale-related information, which is crucial due to the significant scale variations in remote sensing objects.

Second, we performed ablation experiments on various multilevel feature extraction networks, including ResNet-50 [75], Swin-Tiny (Swin-T) [76], HRNet-32 [77], and ViT-S. The input for the pixel decoder requires unifying multilevel features to the same number of channels. For ResNet-50,

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TABLE IV
COMPARISON IN NOC80, NOC85, AND NOC90 BETWEEN DIFFERENT NUMBERS OF CPPD LAYERS AND LTD LAYERS

Number of CPPD	Number of LTD	NWPU			SSDD			WHUB-I			ISPRS Vaihingen		
		NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90
1	3	2.65	3.72	6.59	3.30	6.61	**13.23**	3.29	4.43	7.29	4.17	5.27	7.79
2	3	**2.52**	**3.63**	**6.49**	3.49	7.09	13.55	**3.25**	**4.38**	**7.20**	**3.87**	**4.96**	**7.39**
3	3	2.74	3.86	6.77	3.52	7.03	13.70	**3.25**	4.40	7.21	4.32	5.46	7.84
4	3	2.72	3.87	6.88	**3.21**	**6.47**	**13.23**	3.38	4.60	7.62	4.16	5.25	7.79
2	1	2.65	3.75	6.63	3.80	7.71	13.78	3.27	4.43	7.26	4.15	5.32	7.78
2	2	2.75	3.83	6.76	3.50	7.26	13.90	3.32	4.46	7.32	4.32	5.43	7.89

TABLE V
COMPARISON IN NOC80, NOC85, AND NOC90 BETWEEN USING SINGLE-SCALE FEATURES AND MULTISCALE FEATURES

	NWPU			SSDD			WHUBuilding			ISPRS Vaihingen		
	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90
Single-scale features	3.10	4.46	7.30	3.24	6.44	13.27	3.60	4.89	8.00	4.61	5.82	8.55
Multi-scale features	**2.85**	**4.12**	**6.99**	**3.07**	**6.19**	**12.93**	**3.44**	**4.66**	**7.66**	**4.50**	**5.70**	**8.25**

TABLE VI
TOTAL NUMBER OF PARAMETERS ABOUT MULTILEVEL FEATURE EXTRACTION NETWORKS INCLUDE RESNET-50, SWIN-T, HRNET-32, AND VIT-S

Multi-level Feature Extraction Network	Parameters
ResNet-50 + ChannelMapper	32.36M
Swin-T + ChannelMapper	31.95M
HRNet-32 + ChannelMapper	32.49M
ViT-S + SimpleFPN	25.35M

Swin-T, and HRNet-32, we apply a convolutional layer to each multilevel feature, standardizing the number of channels, called ChannelMapper. ViT-S used a SimpleFPN to obtain multilevel features. All networks except ViT-S utilized pretrained parameters from ImageNet-1k [78]. Their parameter counts are within the same order of magnitude, as shown in Table VI. Training these four multilevel feature extraction networks in the same decoder and methodology yields the results in Table VII.

As shown in Table VII, ViT-S achieved the best results, attributed to its self-attention mechanism, which excels in capturing global features and integrating image and click map features. In addition, pretrained parameters from DINO [73] offer stronger feature extraction capabilities compared with those from ImageNet-1k.

3) CPPD Ablation Experiment: To validate the effectiveness of the click prompt and the ICCAB structure, we conducted three sets of ablation experiments. In the description of Section II, the pixel decoder of current DETR-like segmentation models and interactive segmentation models generally use deformable self-attention, while the method for clicks is a learnable embedding. Combined with the proposed ICCAB, the layer designs for the three sets of ablation experiments are shown in Fig. 12.

In Fig. 12(a), the attention for clicks to the image is removed. In Fig. 12(b), based on (a), the feature embedding is replaced with learnable label embedding. In Fig. 12(c), the pixel decoder layers of the model are all replaced with deformable attention, the same as in [32], [37], and [50]. We conducted experiments on these three structures separately, using ViT-S as the backbone network. The results are shown in Table VIII.

It can be seen that our structure performs the best overall in the ablation experiments for ICCAB. Designs of Fig. 12(a) and (b) perform noticeably worse compared with designs of (c) and ours on the ISPRS Vaihingen dataset. It can be observed that attention only on clicks to the image results in a poor ability to capture the global features of the image. This is because in the ISPRS Vaihingen dataset, there are three categories of noninstance objects, and their class pixels are often distributed across the entire image. The overall performance of design of Fig. 12(a) is better than design of (b). It can be seen that directly utilizing click prompts in the image can achieve better semantic representation compared with learnable embedding, while also reducing the learning parameters of the network. On the NWPU and WHUB-I datasets, designs of Fig. 12(a), (b) and ours outperform (c). It can be seen that using click prompts has certain advantages compared with the commonly used image self-attention methods.

4) LTD Ablation Experiment: For this section, our three main designs are initializing a query, optimizing a query using IoU, and solely utilizing deformable attention. We conducted three sets of ablation experiments to demonstrate the effectiveness of these designs, as shown in Table IX. − Init query represents removing query initialization, replaced with commonly used learnable queries. − IoU prediction represents removing the IoU prediction and replacing it with the category prediction. − Trans. decoder represents removing the Transformer decoder and directly learning a 1×1 convolution to obtain the segmentation results.

The results indicate that query initialization significantly improves performance compared with noninitialized queries. Initialized queries rapidly learn semantic information, providing effective reference points for deformable attention and enhancing global information capture. Transitioning from category prediction to IoU prediction mitigates the limitations

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TABLE VII
COMPARISON IN NOC80, NOC85, AND NOC90 BETWEEN MULTILEVEL FEATURE EXTRACTION NETWORKS INCLUDE RESNET-50, SWIN-T, HRNET-32, AND VIT-S

Multi-level Feature Extraction Networks	NWPU			SSDD			WHUBuilding			ISPRS Vaihingen		
	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90
ResNet-50 + ChannelMapper	3.17	4.68	7.65	3.19	**5.92**	**12.72**	4.03	5.65	9.34	5.29	6.69	9.46
Swin-T + ChannelMapper	3.08	4.41	7.39	3.15	6.50	13.82	3.64	4.99	8.29	4.56	**5.70**	**8.25**
HRNet-32 + ChannelMapper	2.90	4.23	7.18	3.32	6.39	13.52	3.74	5.09	8.32	4.66	5.80	8.26
ViT-S + SimpleFPN	**2.85**	**4.12**	**6.99**	**3.07**	6.19	12.93	**3.44**	**4.66**	**7.66**	**4.50**	**5.70**	**8.25**

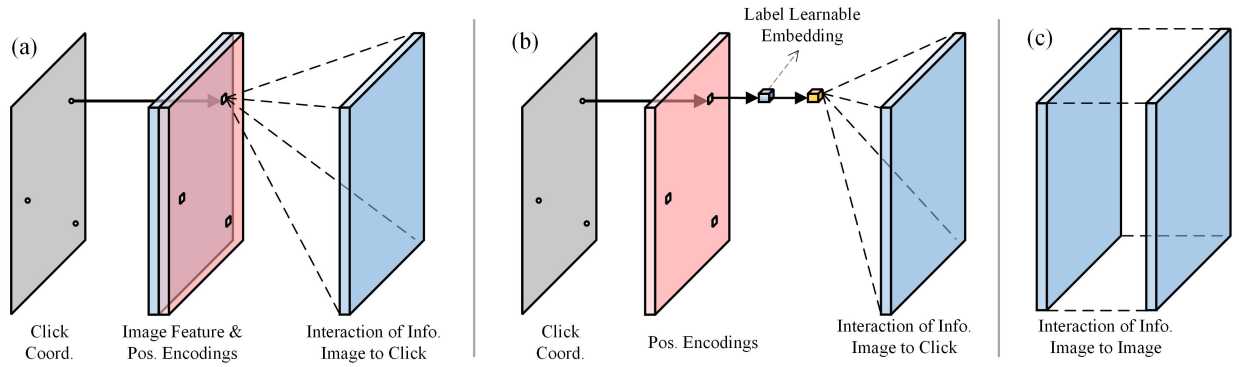


Fig. 12. Design of CPPD ablation experiment. (a) Removes the deformable attention from the click to the image. (b) Replaces the click prompt of the image feature with a label learnable embedding. (c) Removes the attention from the image-to-click and replaces with image self-attention.

TABLE VIII
COMPARISON IN NOC80, NOC85, AND NOC90 BETWEEN DIFFERENT DESIGN CPPD LAYERS

Design	NWPU			SSDD			WHUB-I			ISPRS Vaihingen		
	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90
Ours	**2.85**	**4.12**	**6.99**	3.07	6.19	**12.93**	**3.44**	**4.66**	**7.66**	**4.50**	**5.70**	**8.25**
(a)	3.00	4.24	7.11	3.00	**5.82**	**12.93**	3.52	4.78	7.83	4.98	6.09	8.71
(b)	3.18	4.48	7.32	3.09	6.12	13.04	3.50	4.79	7.87	4.97	6.17	8.64
(c)	3.20	4.52	7.32	**2.99**	5.84	13.03	3.56	4.85	8.00	4.69	5.88	8.54

TABLE IX
COMPARISON IN NOC80, NOC85, AND NOC90 BETWEEN DIFFERENT DESIGN TRANSFORMER DECODER LAYERS

Design	NWPU			SSDD			WHUB-I			ISPRS Vaihingen		
	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90	NoC80	NoC85	NoC90
Ours	**2.85**	**4.12**	**6.99**	3.07	**6.19**	**12.93**	**3.44**	**4.66**	**7.66**	**4.50**	**5.70**	8.25
– Init query	3.00	4.28	7.10	**3.06**	6.21	13.22	3.52	4.77	7.80	4.54	5.72	**8.19**
– IoU prediction	3.08	4.44	7.21	3.10	6.37	13.21	3.47	4.73	7.81	4.71	5.96	8.50
– Trans. decoder	3.06	4.34	7.15	3.09	6.36	13.29	3.53	4.78	7.82	4.66	5.84	8.41

imposed by fewer categories in IIS tasks, enabling more effective global learning. In addition, while SimpleClick surpasses PiClick and AdaptiveClick on remote sensing images, testing whether a direct convolutional layer outperforms a transformer decoder with fewer parameters revealed that the transformer decoder remains essential. Direct learning without a transformer decoder does not achieve the same level of effectiveness in interacting with the image for feature representation.

F. Method Analysis

1) Visualization Analysis of the Click-to-Image Deformable Attention: We visualized the attention location of the click-to-image deformable attention in the two CPPD layers of the pixel decoder, as shown in Fig. 13. The visualization indicates the location of interest in the three-scale features and the attention focuses on after each click, along with the corresponding attention weights.

As shown in Fig. 13, the attention range varies significantly across the three scales. The attention location is primarily concentrated around the click location. $\mathcal{F}_4$ has the widest range of attention, followed by $\mathcal{F}_3$, and $\mathcal{F}_2$ has the smallest. In terms of attention weights, high-weight attention is primarily concentrated around the click location in $\mathcal{F}_2$. For plane and playground, notable high-weight attention positions also appear in $\mathcal{F}_3$ and $\mathcal{F}_4$.

Thus, it can be concluded that in the CPPD layers, the deformable attention primarily concentrates on the regions

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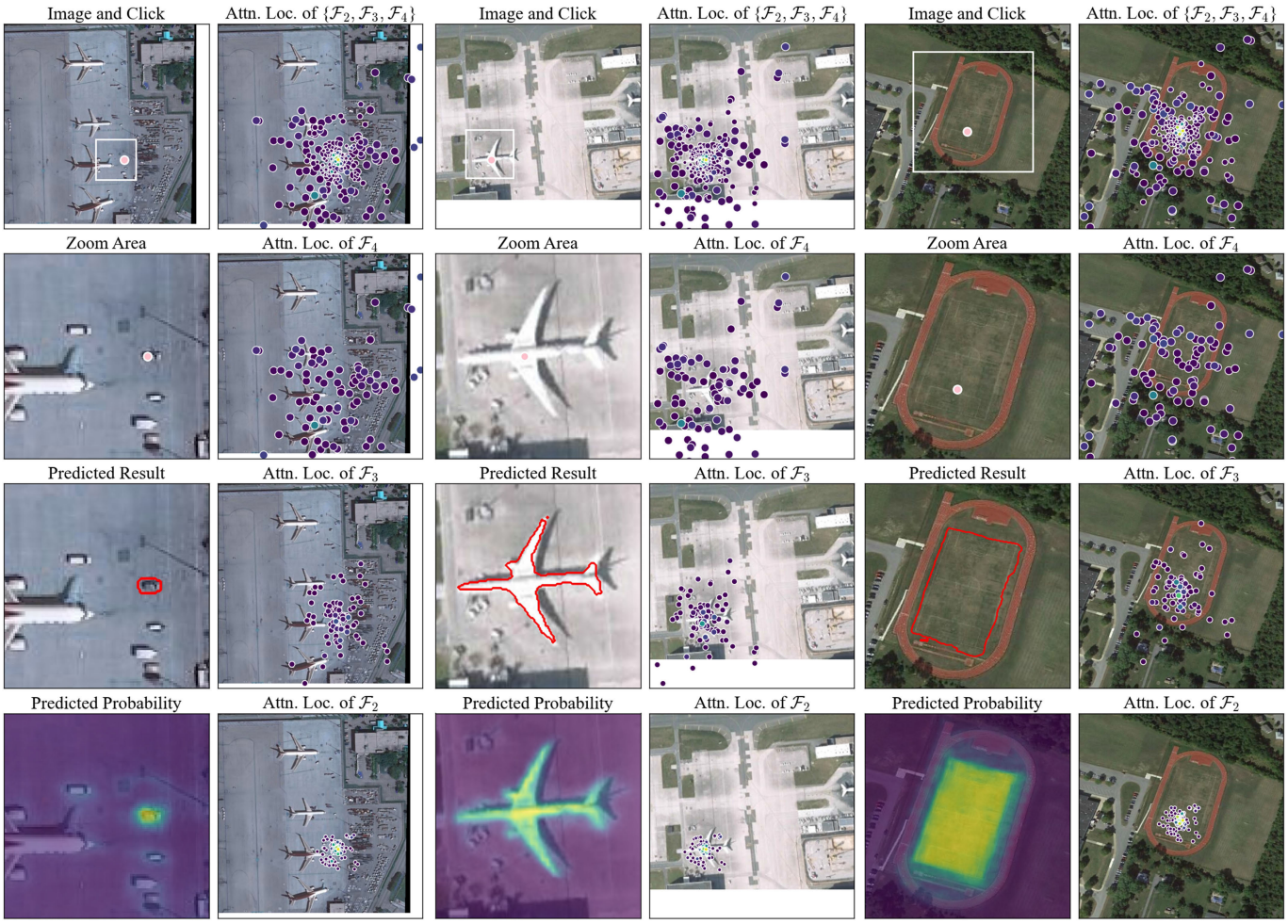


Fig. 13. Attention visualization of the click-to-image deformable attention.

surrounding the click location within the features. And there are different attention ranges on $\mathcal{F}_2$, $\mathcal{F}_3$, and $\mathcal{F}_4$. The high attention weight is mainly in $\mathcal{F}_2$. As the target scale increases, the weights will increase in $\mathcal{F}_3$ and $\mathcal{F}_4$.

2) IoU Prediction Analysis of the Transformer Decoder: We performed simulated click tests on each object across the four remote sensing datasets. With a target IoU of 0.9 and a maximum of 20 clicks, we recorded the predicted IoU and the actual IoU throughout the interaction process. The residuals between the predicted and actual IoU were computed and are visualized in Fig. 14, with data indices on the x-axis and the residual on the y-axis.

It can be observed that there is a notable discrepancy between the predicted IoU and the actual IoU. First, during simulated testing, the predicted results following a click often correspond to two overlapping objects located in the same position within the dataset, resulting in larger residuals, particularly in cases involving object inclusion relationships. Second, the single-category dataset structure limits the diversity of training scenarios, hindering the model's IoU prediction branch from achieving optimal performance.

Nevertheless, compared with category prediction, IoU prediction encourages the model to learn richer feature representations, ultimately resulting in more accurate annotations.

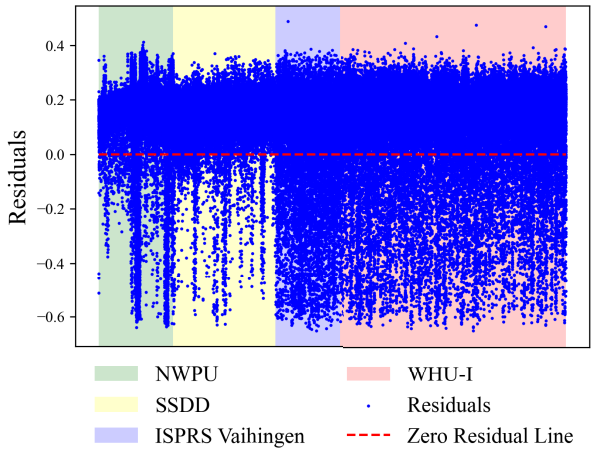


Fig. 14. Residual between the predicted IoU and the ground-truth IoU.

3) Method's Generalizability: To evaluate the generalizability of our method, we tested its performance alongside several methods on natural image datasets, including GrabCut [42], Berkeley [79], DAVIS [80], PascalVOC [81], and SBD [82], as well as medical image datasets such as BraTS [83],

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TABLE X
COMPARISON IN NOC85 AND NOC90 IN NATURAL IMAGES DATASETS OF GRABCUT, BERKELEY, DAVIS, PASCALVOC, AND SBD

Method		GrabCut	Berkeley	DAVIS	VOC	SBD
RITM	NoC85	4.94	6.10	10.62	5.22	7.25
	NoC90	5.98	8.42	13.14	6.28	10.17
SimpleClick	NoC85	3.04	**3.73**	8.42	4.26	6.08
	NoC90	3.52	5.29	10.34	5.08	8.65
AdaptiveClick	NoC85	3.32	4.16	8.53	4.53	6.43
	NoC90	3.80	5.72	10.52	5.41	9.09
Ours	NoC85	**2.84**	3.81	**6.40**	**4.00**	**5.78**
	NoC90	**3.22**	**4.94**	**8.31**	**4.76**	**8.29**

TABLE XI
COMPARISON IN NOC85 AND NOC90 IN MEDICAL IMAGES DATASETS OF BRATS, SSTEM, AND OAIZIB

Method		BraTS	ssTEM	OAIZIB
RITM	NoC85	8.46	2.43	17.98
	NoC90	12.10	4.07	19.89
SimpleClick	NoC85	6.88	2.56	17.11
	NoC90	10.82	3.84	19.85
AdaptiveClick	NoC85	7.24	2.57	**15.64**
	NoC90	11.34	3.51	**19.37**
Ours	NoC85	**6.56**	**2.42**	17.16
	NoC90	**10.57**	**3.38**	19.78

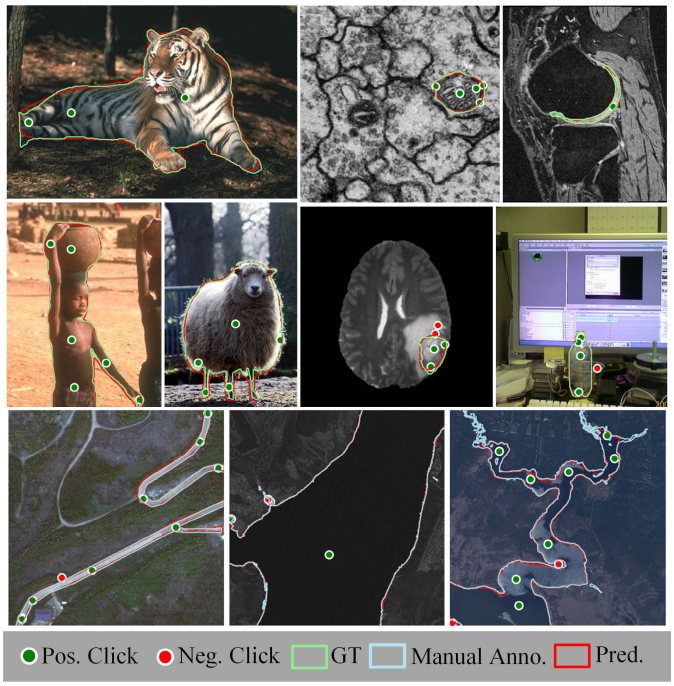


Fig. 15. Our method interactive segmentation on natural images, medical images, and remote sensing images.

ssTEM [84], and OAIZIB [85]. The selected comparison methods include RITM, the best-performing CNN-based model; SimpleClick, the leading Transformer-based model; and AdaptiveClick, the top-performing DETR-like method.

The results for the natural image datasets are presented in Table X, and the results for the medical image datasets are shown in Table X.

From Tables X and XI, it can be observed that the annotation model trained on remote sensing images exhibits a certain degree of generalizability to both natural and medical images. In the IIS task, the model primarily learns semantic and edge-related information. When applied to nonremote sensing images, the model relies mainly on edge information for object annotation. This enables good performance on natural images with distinct edges, as well as certain medical images. However, the OAIZIB dataset, which consists of knee-related MR images, introduces significant noise and contains semantic information that the model has not been trained to recognize, leading to reduced performance.

Fig. 15 illustrates the segmentation results for datasets from different domains. When provided with an adequate NoC, the model typically performs well in segmenting simple objects. However, for objects with highly complex edge structures, the model trained on remote sensing datasets exhibits difficulties in achieving effective segmentation. At the same time, Fig. 15 also presented a comparison between the results of manual annotation and the annotation results of this method in the actual remote sensing objects labeling process. Our method was not trained on the related dataset. It can be seen that this method can quickly segment the road and water at the corresponding scale. But, in the small tributary of the river, it is still necessary to manually adjust the scale for fine segmentation.

V. CONCLUSION

In this article, we proposed a DETR-like network of click prompt IIS on high-resolution remote sensing images, redesigned the pixel decoder and the transformer decoder within the network. In the pixel decoder, we designed a method that allows click prompts to learn label information, semantic information, and positional information. We utilized this click prompt to establish mutual attention with the image. This allows the image features to concentrate on the annotated clicks and their surrounding pixels, thereby enhancing the segmentation results. In the transformer decoder, we simplified the attention module by directly employing deformable attention to enable the query to acquire global information. In addition, by initializing the query and optimizing IoU loss, the query can capture global information more swiftly and effectively. The method was trained and tested on the iSAID, ISPRS Vaihingen, ISPRS Potsdam, WHU Building, NWPU, and SSDD datasets. These datasets cover most land cover classes under high-resolution remote sensing images and demonstrate adaptability to nontraining data domains. Furthermore, our method achieved the best quantitative results compared with current state-of-the-art methods. We can obtain more accurate annotations with fewer clicks, and our model is better suited for remote sensing images compared with existing natural image pretraining models like SAM. In addition, we conducted ablation experiments on the design of the pixel decoder and transformer decoder, demonstrating the effectiveness of the structural design. We will further explore the integration of interactive content with IIS, expanding the forms of interactive content to achieve a more convenient interactive experience and more accurate annotation results.

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